

A Counterexample to the Milman-Moduli Quasi-Orthogonality Conjecture

Run fa_banach_001, agent_lane_10

June 16, 2026

Source question

The source is Grigory Ivanov and Horst Martini, *New Moduli for Banach Spaces*, arXiv:1609.01587. In Section 6, after defining Milman's moduli

$$\beta_X^-(\varepsilon) = \inf_{x,y \in S_X} (\max\{\|x + \varepsilon y\|, \|x - \varepsilon y\|\} - 1)$$

and

$$\beta_X^+(\varepsilon) = \sup_{x,y \in S_X} (\min\{\|x + \varepsilon y\|, \|x - \varepsilon y\|\} - 1),$$

the authors conjecture that it is enough to restrict to pairs with y quasi-orthogonal to x . With their notation

$$\zeta_X^-(\varepsilon) = \inf\{\|x + \varepsilon y\| : x, y \in S_X, y \perp x\}, \quad \zeta_X^+(\varepsilon) = \sup\{\|x + \varepsilon y\| : x, y \in S_X, y \perp x\},$$

the conjectured identities are

$$\zeta_X^-(\varepsilon) - 1 = \beta_X^-(\varepsilon), \quad \zeta_X^+(\varepsilon) - 1 = \beta_X^+(\varepsilon)$$

for positive ε . Here $y \perp x$ means that some $p \in J_1(x)$ satisfies $p(y) = 0$.

The definitions of the moduli $\zeta_X^-(\cdot) - 1$ and $\zeta_X^+(\cdot) - 1$ are similar to the definitions of Milman's moduli, which were introduced in [22] as

$$\beta_X^-(\varepsilon) = \inf_{x,y \in \partial \mathfrak{B}_1(o)} \{\max\{\|x + \varepsilon y\|, \|x - \varepsilon y\|\} - 1\}$$

and

$$\beta_X^+(\varepsilon) = \sup_{x,y \in \partial \mathfrak{B}_1(o)} \{\min\{\|x + \varepsilon y\|, \|x - \varepsilon y\|\} - 1\}.$$

We think that in the definitions of Milman's moduli it is sufficient to take only $y \perp x$. Hence we get

Conjecture 3. *Let X be an arbitrary Banach space. Then for positive ε we have*

$$\zeta_X^-(\varepsilon) - 1 = \beta_X^-(\varepsilon) \text{ and } \zeta_X^+(\varepsilon) - 1 = \beta_X^+(\varepsilon).$$

REFERENCES

- [1] J. Alonso and C. Benitez. Some characteristic and non-characteristic properties of inner product spaces. *J. Approx. Theory*, 55:318–325, 1988.
- [2] J. Alonso, H. Martini, and M. Spirova. Minimal enclosing discs, circumcircles, and circumcenters in normed planes. II. *Comm. Geom.*, 45(7):350–369, 2012.

Result

Theorem 1. *The first identity in the conjecture is false. In the real Banach space $X = \ell_4^2$,*

$$\zeta_X^-(1/2) - 1 < \beta_X^-(1/2).$$

Idea of the proof

The quantity β_X^- is protected by a two-sign maximum: for every pair (u, v) , at least one of $\|u + \varepsilon v\|$ and $\|u - \varepsilon v\|$ must be large. The quantity ζ_X^- only asks about the single chosen sign $\|x + \varepsilon y\|$, even when y lies in a supporting hyperplane at x . In ℓ_4^2 , a point x very close to a coordinate vector has a tangent direction whose negative second coordinate moves slightly inward for the positive sign $x + \varepsilon y$. This lowers the one-sign value below the two-sign Milman lower bound.

Proof

Work in $X = (\mathbb{R}^2, \|\cdot\|_4)$, where

$$\|(a, b)\|_4 = (|a|^4 + |b|^4)^{1/4}.$$

Lemma 1. *For every $\varepsilon > 0$,*

$$\beta_{\ell_4^2}^-(\varepsilon) + 1 = (1 + \varepsilon^4)^{1/4}.$$

Proof. For real numbers a, b ,

$$|a + b|^4 + |a - b|^4 = 2(a^4 + 6a^2b^2 + b^4) \geq 2(a^4 + b^4).$$

Hence, if $u, v \in S_{\ell_4^2}$, then

$$\|u + \varepsilon v\|_4^4 + \|u - \varepsilon v\|_4^4 = 2(1 + \varepsilon^4) + 12\varepsilon^2 \sum_i u_i^2 v_i^2 \geq 2(1 + \varepsilon^4).$$

Therefore

$$\max\{\|u + \varepsilon v\|_4, \|u - \varepsilon v\|_4\} \geq (1 + \varepsilon^4)^{1/4}.$$

Equality is attained by $u = (1, 0)$ and $v = (0, 1)$, since both $\|u + \varepsilon v\|_4$ and $\|u - \varepsilon v\|_4$ equal $(1 + \varepsilon^4)^{1/4}$. $\square$

It remains to exhibit one quasi-orthogonal pair with a smaller one-sign value at $\varepsilon = 1/2$. Put

$$s = \frac{1}{6}, \quad a = (1 - s^4)^{1/4}, \quad x = (a, s).$$

Then $\|x\|_4 = 1$. Let

$$r = \left(\frac{s}{a}\right)^3, \quad c = (1 + r^4)^{-1/4}, \quad y = c(r, -1).$$

Then $\|y\|_4 = 1$. The norming functional at x is

$$p = (a^3, s^3) \in (\ell_4^2)^* = \ell_{4/3}^2,$$

because $\|p\|_{4/3} = 1$ and $p(x) = a^4 + s^4 = 1$. Moreover

$$p(y) = c(a^3r - s^3) = 0.$$

Thus $y^\top x$ in the convention of the source paper.

We now bound the value of $\|x + \frac{1}{2}y\|_4$. Since

$$a^4 = 1 - \frac{1}{6^4} = \frac{1295}{1296} > \left(\frac{5}{6}\right)^4,$$

we have $a > 5/6$, and hence

$$0 < r = \left(\frac{s}{a}\right)^3 < \left(\frac{1/6}{5/6}\right)^3 = \frac{1}{125}.$$

Also $0 < c < 1$. Therefore the first coordinate of $x + \frac{1}{2}y$ satisfies

$$0 < a + \frac{cr}{2} < 1 + \frac{1}{250} = \frac{251}{250},$$

and the second coordinate satisfies

$$\left|s - \frac{c}{2}\right| < \frac{1}{3}.$$

Consequently

$$\|x + \frac{1}{2}y\|_4^4 < \left(\frac{251}{250}\right)^4 + \left(\frac{1}{3}\right)^4 = \frac{325405456081}{316406250000}.$$

The last rational number is strictly smaller than $17/16$, since

$$\frac{17}{16} - \frac{325405456081}{316406250000} = \frac{673511534}{19775390625} > 0.$$

Thus

$$\zeta_{\ell_4^2}^-(1/2) \leq \|x + \frac{1}{2}y\|_4 < \left(\frac{17}{16}\right)^{1/4} = \beta_{\ell_4^2}^-(1/2) + 1.$$

Subtracting 1 gives

$$\zeta_{\ell_4^2}^-(1/2) - 1 < \beta_{\ell_4^2}^-(1/2),$$

as claimed.

Verification notes

The companion script `code/check_14.counterexample.py` verifies the rational inequality above and prints high-precision numerical values. The actual explicit pair gives approximately

$$\|x + \frac{1}{2}y\|_4 = 1.0051760632186776, \quad (17/16)^{1/4} = 1.0152715924344654.$$

Novelty and literature check

The following bounded searches were performed before promotion: the run's registry, solutions, attempts, and proof-gap indexes were searched for 1609.01587, the title **New Moduli for Banach Spaces**, Milman moduli, $\beta^\pm$, $\zeta^\pm$, quasi-orthogonality, Birkhoff-James orthogonality, and related keywords. Web searches were also run for the exact paper title, 1609.01587, Milman's moduli, **zeta beta Banach**, and the quasi-orthogonality phrasing of the conjecture. No later answer to Conjecture 3 was found in this bounded search; the only direct hits were the source paper and generic bibliographic entries.

Limitations

This packet disproves only the equality

$$\zeta_X^-(\varepsilon) - 1 = \beta_X^-(\varepsilon)$$

from Conjecture 3, by the single example $X = \ell_4^2$ and $\varepsilon = 1/2$. It does not decide the proposed equality involving ζ_X^+ and β_X^+ , nor the other questions and conjectures in the source paper.

Human-review recommendation

This is a short, explicit counterexample with exact rational inequalities. Human review should mainly confirm the source convention for quasi-orthogonality and the elementary computation of $\beta_{\ell_4^2}^-$.

References

- [1] G. Ivanov and H. Martini, *New Moduli for Banach Spaces*, arXiv:1609.01587.
- [2] V. D. Milman, *Geometric theory of Banach spaces. Part II. Geometry of the unit ball*, Russian Mathematical Surveys 26 (1971), 79–163.